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superblockify: A Python Package for Automated Generation, Visualization, and Analysis of Potential Superblocks in Cities

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Summary

superblockify is a Python package for partitioning an urban street network into Superblock-like neighborhoods and for visualizing and analyzing the partition results. A Superblock is a set of adjacent urban blocks where vehicular through traffic is prevented or pacified, giving priority to people walking and cycling (Nieuwenhuijsen et al., 2024). The Superblock blueprints and descriptive statistics generated by superblockify can be used by urban planners as a first step in a data-driven planning pipeline, or by urban data scientists as an efficient computational method to evaluate Superblock partitions. The software is licensed under AGPLv3 and is available at https://superblockify.city.

Statement of need

The Superblock model is an urban planning intervention with massive public health benefits that creates more liveable and sustainable cities (Laverty et al., 2021; Mueller et al., 2020; WHO, 2022). Superblocks form human-centric neighborhoods with reduced vehicular traffic. They are safer, quieter, and more environmentally friendly (Agència d'Ecologia Urbana de Barcelona et al., 2021; Martin, 2021; Mueller et al., 2020) than car-centric urban landscapes which fully expose citizens to car harm (Miner et al., 2024). The scientific study of Superblocks has expanded quickly in recent years, summarized in a review by Nieuwenhuijsen et al. (2024). The planning and implementation of Superblocks is an intricate process, requiring extensive stakeholder involvement and careful consideration of trade-offs (Nieuwenhuijsen et al., 2019; Stadt Wien, 2021; Transport for London, 2020). New computational tools and data sets, such as the osmnx Python library (Boeing, 2017) and OpenStreetMap (OpenStreetMap contributors, 2023), provide the opportunity to simplify this process by allowing to easily analyze and visualize urban street networks computationally. Recent quantitative studies on Superblocks have seized this opportunity with different focuses, such as potential Superblock detection via network flow on the abstract level (Eggimann, 2022a) or in the local context of Vienna (Frey et al., 2020); development of interactive micro-level planning tools (Carlino et al., 2024; *TuneOurBlock*, 2024); green space (Eggimann, 2022b), social factors (Yan & Dennett, 2023), health benefit modeling (Li & Wilson, 2023), or an algorithmic taxonomy of designs (Feng & Peponis, 2022). However, to our knowledge, none of these emerging research efforts have led to an open, general-use, extendable software package for Superblock delineation, visualization, and analysis. superblockify fills this gap.

The software offers benefits for at least two use cases. First, for urban planning, it provides

a quick way to generate Superblock blueprints for a city, together with descriptive statistics informing the planning process. These blueprints can serve as a vision or first draft for potential future city development. In a planning pipeline, `superblockify` stands at the beginning, broadly delineating the potential areas of study first. Then, exported Superblocks can feed into an open geographic information system like QGIS (QGIS Development Team, 2024) or into tools like A/B Street (Carlino et al., 2024) or TuneOurBlock (*TuneOurBlock*, 2024) that allow finetuned modifications or traffic simulations. This quick feedback can reduce the time and resources required to manually plan Superblocks, which in turn can accelerate sustainable urban development. Second, `superblockify` enables researchers to conduct large-scale studies across multiple cities or regions, providing valuable insights into the potential impacts of Superblocks at a broader scale. In both cases, `superblockify` can help to identify best practices, algorithmic approaches, and strategies for Superblock implementation.

The software has served in a preliminary analysis of potential Superblocks in 180 cities worldwide (Büth, 2023) and will be used in subsequent studies within the EU Horizon Project JUST STREETS (https://just-streets.eu). With increased urbanization, impacts of climate change, and focus on reducing car-dependence (Mattioli et al., 2020; Ritchie & Roser, 2018; Satterthwaite, 2009), the need for sustainable urban planning tools like `superblockify` will only increase (Nieuwenhuijsen et al., 2024).

Features

`superblockify` has three main features: Data access and partitioning, Visualization, and Analysis.

Data access and partitioning

`superblockify` leverages OpenStreetMap data (OpenStreetMap contributors, 2023) and population data GHS-POP R2023A (Pesaresi & Politis, 2023). From a user-given search query, e.g., a city name, `superblockify` retrieves the street network data of a city, the necessary GHS-POP tile(s), and distributes the population data onto a tesselation of the street network.

After the street network and optional metadata are loaded in, the package partitions the street network into Superblocks. In its current version 1.0.0, `superblockify` comes with two partitioners:

1. The residential approach uses the given residential street tag to decompose the street network into Superblocks.
2. The betweenness approach uses the streets with high betweenness centrality for the decomposition.

The resulting Superblocks can be exported in GeoPackage (`.gpkg`) format for further use.

Visualization

After the partitioning, factors relevant for analysis and planning of Superblocks can be calculated and visualized, e.g., area, population, population density, or demand change by betweenness centrality. Example Superblock configurations for two cities are shown in Fig. 1.

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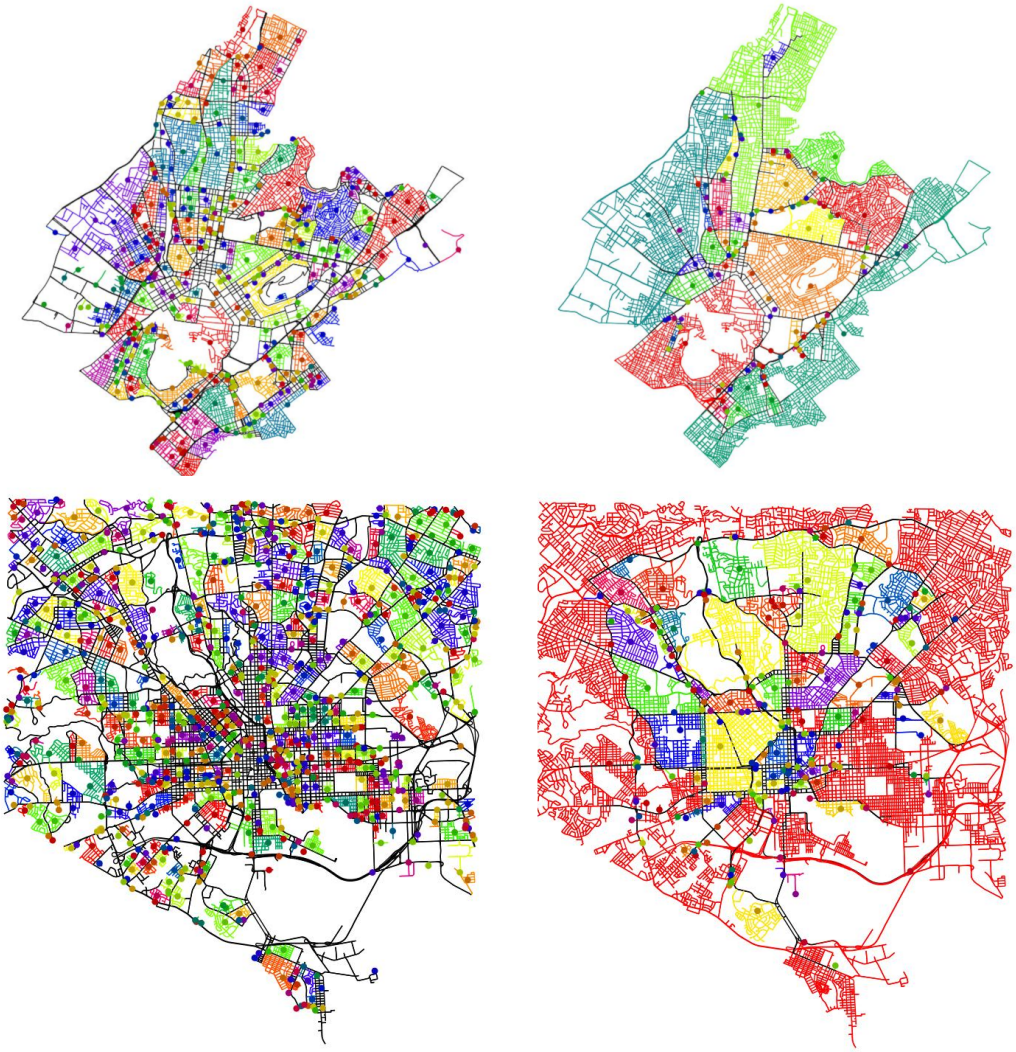

Figure 1: Automated generation of Superblocks. Athens (top row) and Baltimore (bottom row) Superblocks generated using the residential partitioner (left column) and the betweenness partitioner (right column). The streets of each Superblock are colored, the rest of the streets are black. Colored nodes denote representative nodes within each Superblock for easier visual recognition. Map data from OpenStreetMap.

Analysis

For analysis, the package calculates various graph metrics of the street network, such as global efficiency (Latora & Marchiori, 2001), directness (Szell et al., 2022), betweenness centrality (Brandes, 2008), spatial clustering and anisotropy of high betweenness centrality nodes (Kirkley et al., 2018), street orientation-order (Boeing, 2019b), or average circuity (Boeing, 2019a). These metrics are calculated for the entire street network and for each Superblock individually. To facilitate further analysis, all of these metrics are included in the exportable GeoPackage file.

Design

`superblockify`'s design is object-oriented with a focus on modularity and extensibility. An abstract partitioner base class is provided to facilitate implementing new custom approaches for

Superblock generation. At the core of the package, `superblockify` extends Dijkstra's efficient distance calculation approach with Fibonacci heaps on reduced graphs, ensuring optimal performance when iterating various Superblock configurations while respecting the Superblock restriction of no through traffic. This restriction is checked via just-in-time (JIT) compilation through `numba` (Lam et al., 2023) to speed up the calculation of betweenness centrality on directed, large-scale street networks. Central code dependencies are the `osmnx` (Boeing, 2017) and `networkx` (Hagberg et al., 2008) packages for data acquisition, preprocessing, and network analysis, and the geopandas (Bossche et al., 2023) package for spatial analysis.

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Authors contributions with CRediT

- Carlson M. Büth: Conceptualization, Software, Investigation, Methodology, Writing – original draft, Validation
- Anastassia Vybornova: Conceptualization, Supervision, Writing – review & editing, Validation
- Michael Szell: Conceptualization, Project administration, Writing – review & editing, Validation, Funding acquisition

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